

Duolingo App Review Analysis

By Shikha Punjabi

Abstract

In this project, I analyzed over 2,000 real user reviews from the Duolingo Android app to uncover sentiment patterns, identify user pain points, and simulate the potential impact of product improvements. By applying techniques from natural language processing (NLP), classification modeling, and business analysis, I explored how features like 'hearts', 'ads', and 'streaks' influence user sentiment and app retention. This project bridges data science with strategic product recommen...

Introduction

Duolingo, a leader in gamified language learning, has built a reputation around habit-forming engagement strategies like streaks, leaderboards, and progress rewards. While this gamification encourages frequent use, it can also introduce friction that negatively affects the user experience.

This project began with a business question: what exactly do users enjoy, and what frustrates them enough to stop using the app?

To answer that, I applied a business analyst's lens-framing sentiment analysis, trend detection, and predictive modeling in terms of customer retention, experience design, and product optimization. The result is a data-backed strategy proposal for enhancing engagement and reducing churn.

Methodology Overview

- Data Source: ~2,200 reviews pulled using the 'google_play_scraper' library (English language, global)
- Preprocessing: Removed short/duplicate entries, applied regex cleaning, lemmatization, and stopword filtering
- Feature Tagging: Defined keyword-based feature mappings to tag mentions of "hearts", "ads", "streaks", etc.
- Sentiment Analysis: Used VADER to classify polarity scores and label sentiments (positive, neutral, negative)
- Churn Risk Modeling: Trained a logistic regression model to classify reviews as likely to indicate churn or not
- Forecasting: Built a time-series model using Prophet to predict sentiment trends over time
- Simulation: Modeled a scenario where the "hearts" feature is relaxed or removed to quantify its impact on churn

Key Insights

- 'Hearts' consistently generated frustration. Nearly 45% of reviews mentioning it were negative. Users described it as punitive and demotivating when they made mistakes.
- In a simulated scenario where the 'hearts' system is removed, predicted churn dropped from 46% to 29%.
- Gamification features like 'streaks' and 'leaderboards' produce mixed feedback - some users find them motivating, others stressful or anxiety-inducing.
- Sentiment varies greatly by feature. A heatmap of average sentiment showed polarizing results, helping product managers prioritize which features to tweak or defend.

Recommendations

1. A/B Test a 'Heartless' Learning Mode: Create a test group that uses Duolingo without the heart system. Compare session lengths, retention rates, and sentiment metrics.
2. Customize Gamification: Offer toggle settings for streak tracking or leaderboard participation, improving inclusivity for casual learners.
3. Create Feedback Loops for Product Monitoring: Deploy sentiment dashboards internally using similar NLP tagging pipelines to track review changes post-update.

Deliverables

- Jupyter Notebooks: EDA, sentiment modeling, churn simulation
- Python Scripts: Feature tagging, preprocessing, forecasting
- Streamlit Dashboard: <https://duolingo-analysis.streamlit.app>
- GitHub Repository: <https://github.com/shikha-punjabi/duolingo-analysis>

Conclusion

This project demonstrates how unstructured user feedback can be transformed into actionable business strategy. Duolingo's app is beloved for its creativity and gamification, but it also shows that poorly-tuned mechanics can cause dissatisfaction and drive users away.

By simulating product changes and forecasting user sentiment, this analysis supports proactive decision-making for product teams. It also highlights my ability to extract, process, and communicate insights - translating raw data into product-level impact.